

Introduction & Hook

Ask students:

“Why can’t we always use ultra-high detailed 3D models in VR?”

Explain:

VR requires:

- High FPS (60–120 FPS)
- Low latency

Too many polygons = slow rendering = motion sickness

Today we study how geometry is managed efficiently.

Meshes and Polygons

What is a Polygon?

A polygon is:

A flat surface made of vertices connected by edges.

In 3D graphics:

- Most polygons are triangles
- Triangle = simplest stable 3D shape

Example:

A cube = 12 triangles

Explain:

All 3D models in VR are made of triangles.

What is a Mesh?

Definition:

A mesh is a collection of vertices, edges, and polygons that form a 3D object.

A mesh contains:

- Geometry data
- Texture coordinates
- Normal vectors

Example:

Car mesh = 50,000 polygons

Photogrammetry statue = 2 million polygons

Explain:

Higher polygon count → more detail → more GPU cost.

PART 2: Tessellated Data



25–35 min → What is Tessellation?

Definition:

Tessellation is the process of dividing a surface into smaller polygons (usually triangles).

It increases surface detail.

Example:

A flat terrain → tessellated into many small triangles → smoother appearance.

Why Tessellation is Used

- Improve realism
- Create smooth curves
- Enhance detail close to camera

Problem:

More tessellation =

- More vertices
- More GPU computation
- Lower FPS

So we must balance realism and performance.

PART 3: Level of Detail (LOD)

What is LOD?

Definition:

LOD (Level of Detail) is a technique where different versions of a model are used depending on distance from the viewer.

Example:

Distance	Model Used
Very close	High polygon
Medium	Medium polygon
Far away	Low polygon

User cannot see fine detail at distance, so high detail is unnecessary.

Types of LOD Techniques

1. Discrete LOD

Switch between separate models.

2. Continuous LOD

Gradual reduction of detail.

3. Automatic LOD (Engine-based)

Unity/Unreal auto-generate simplified meshes.

4. Distance-based LOD

Switches based on camera distance.

Tessellation vs LOD (Clarification)

Tessellation	LOD
Increases detail	Reduces detail when needed
Used for realism	Used for performance
Adds polygons	Reduces polygons
Heavy GPU load	Saves GPU load

Important teaching line:

Tessellation adds detail.
LOD removes unnecessary detail.

Real VR Example

In a VR city:

- Nearby building → 100,000 polygons
- Far building → 5,000 polygons
- Mountains → very low poly

Without LOD:

- GPU overload
- Frame drops
- Motion sickness

Recap + Questions

Recap:

- Mesh = collection of polygons
- Tessellation divides surfaces into smaller polygons
- LOD reduces complexity based on distance
- Both affect VR performance

Quick Oral Questions:

- Why are triangles preferred in 3D graphics?
- What is tessellation?
- Why is LOD critical in VR?
- Does tessellation increase or decrease polygons?